MIL-C-24643/37B
3 October 1986
SUPERSEDING
MIL-C-0024643/37A(SH)
8 November 1985
MIL-C-24643/37
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3U (INCLUDING VARIATION LS3UA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS3U (unarmored) and LS3UA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.023 inch. Colored insulation - one conductor black, one conductor white and one conductor red for each triad.
Fourth - One black, one white and one red conductor cabled together with a nominal lay of 3-1/2 inches to form a triad.
Fifth - Marker braid on each triad. Standard identification code applied by method 4. Diameter over braided triad 0.215 inch nominal.
Sixth - The specified number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad no. 1, from center outward. Fillers as necessary to form a firm, round assembly.
Seventh - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS3U

Eighth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS3UA

Eighth - Cross-linked polyolefin jacket. (See table I for thickness).
Ninth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/37	Size 2/	Number of triads	Cable jacket thickness nominal (inch)	Overall diameter (inches)			
				Type LS3U		Type LS3UA	
				minimum	maximum	minimum	maximum
-01UN	3	3	0.040	0.580	0.620	0.630	0.670
-02UN	7	7	.040	.760	.810	.810	.860
-03UN	12	12	.050	1.030	1.090	1.080	1.140

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS3U and LS3UA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Capacitance	
Mutual capacitance - at 1 kHz, pF/ft, maximum.....	30
Capacitance unbalance - percent, maximum.....	5
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LS3U only - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor - conformance to material, construction and coverage (type LS3UA only)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:
All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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